Team Members

Francisco Baptista - 124677
Inês Batista - 124877
Marcos Costa - 125882
Maria Quinteiro - 124996
Luís Correia - 125264

Advisors

João Almeida
Vicente Barros
Sebastião Teixeira

PEI 26/27 - Group 2 - 3rd November 2026

TreeLens

Say Goodbye to Field Surveys

“TreeLens is a gamified, community-driven app that maps a city’s trees”

Who counts the trees in your city?

01: The need

A tree inventory by law

Cities must know which trees they have, where, and in what state and condition.

→

02: The gap

The data does not exist

No dataset of labelled, geolocated urban trees fits our needs.

→

03: The ideia

Let the community collect it

People photograph and catalogue trees. It should be noted that this does not rule out the existence of validation prior to integration into the system.

Actors



Field Surveyor



Municipal Technician

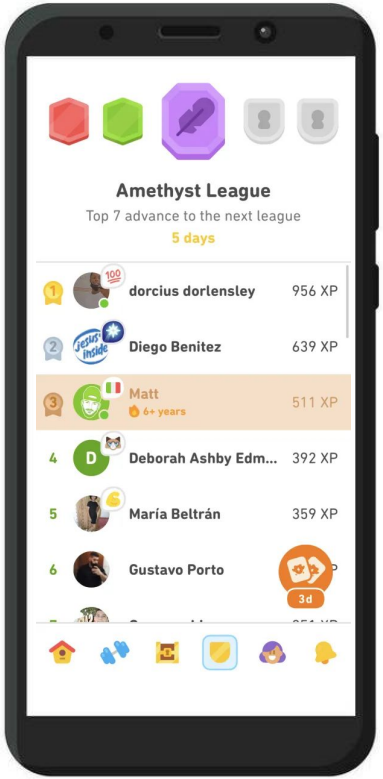


Municipal Administrator



Tree Enthusiast

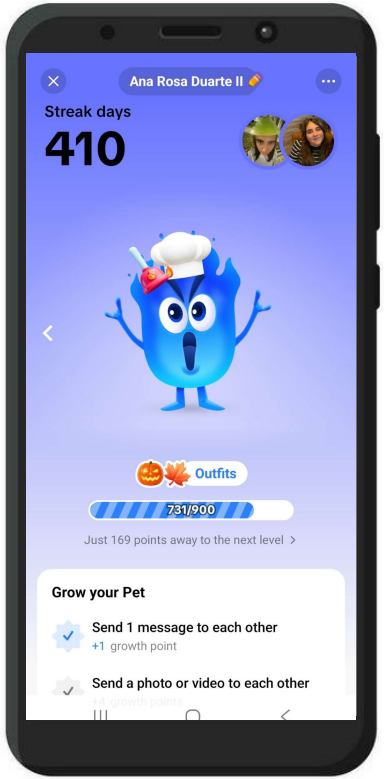
Gamification



Leaderboards from Duolingo

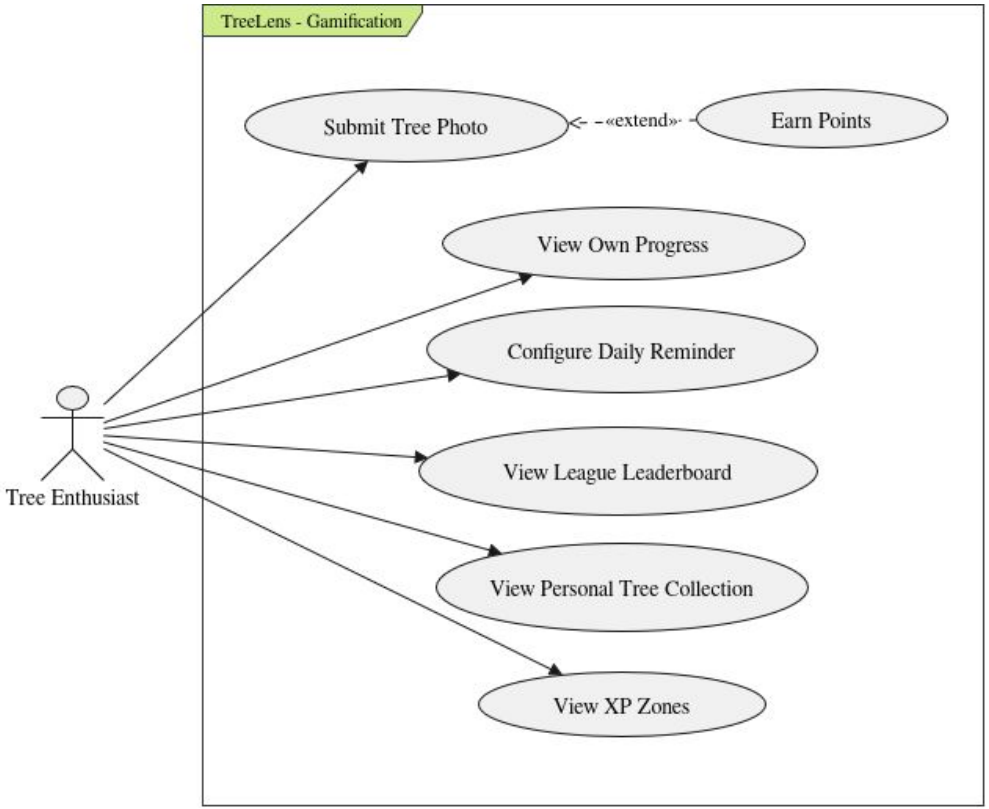


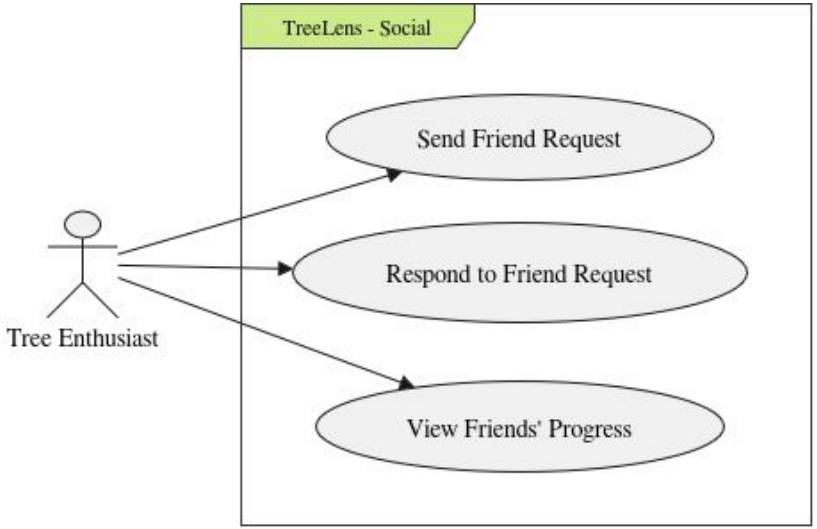
PokeDex from Pokemon GO

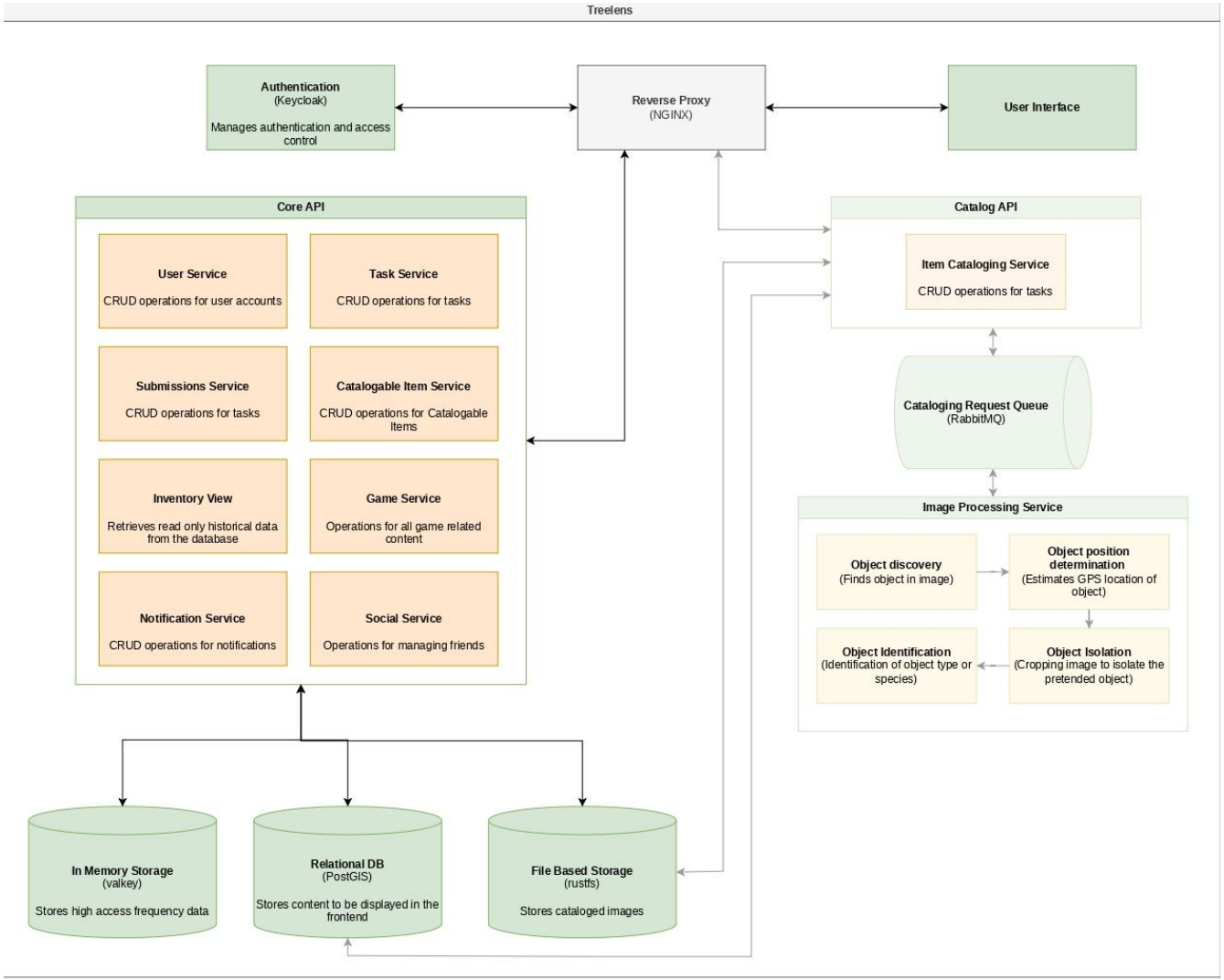


Streak Mascot from TikTok

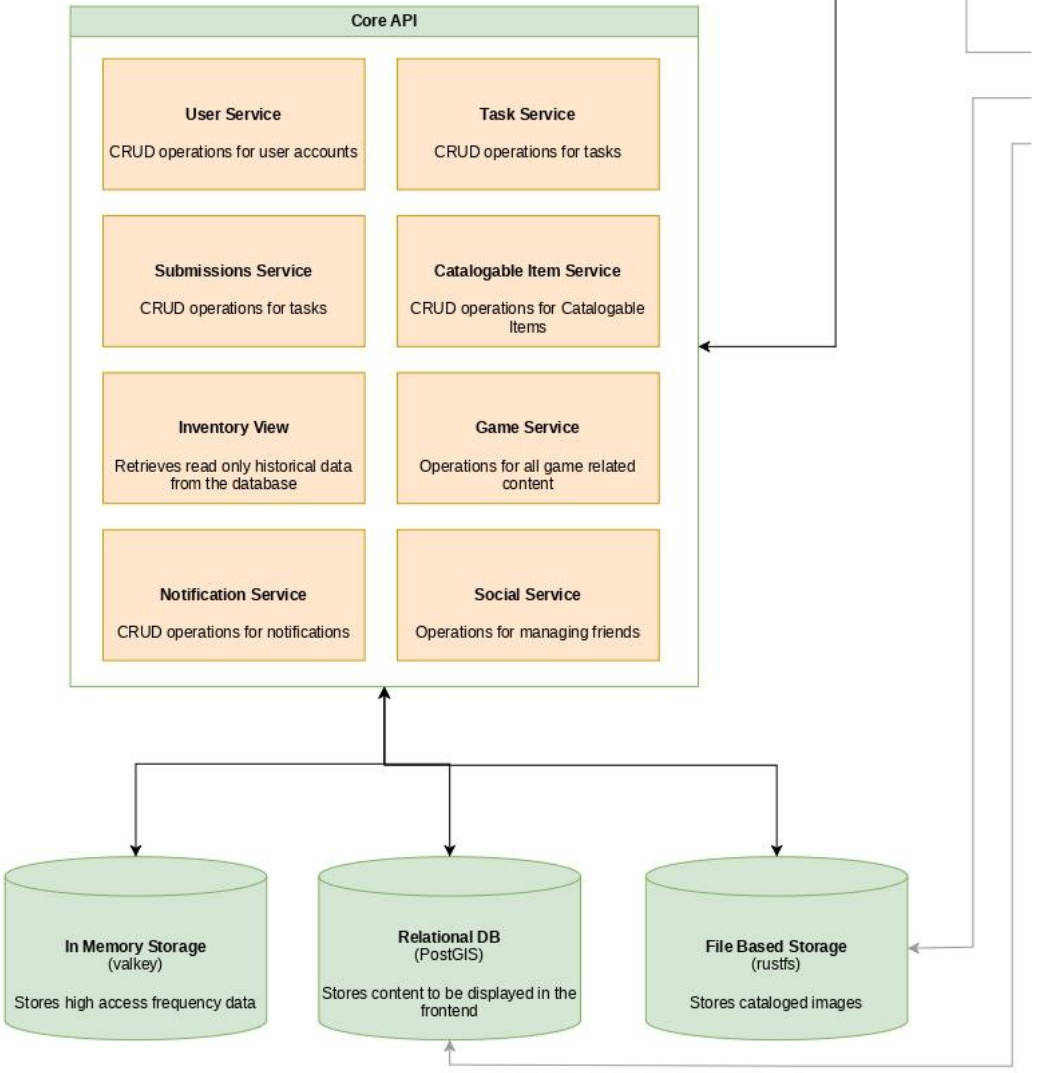
Use Cases Diagrams



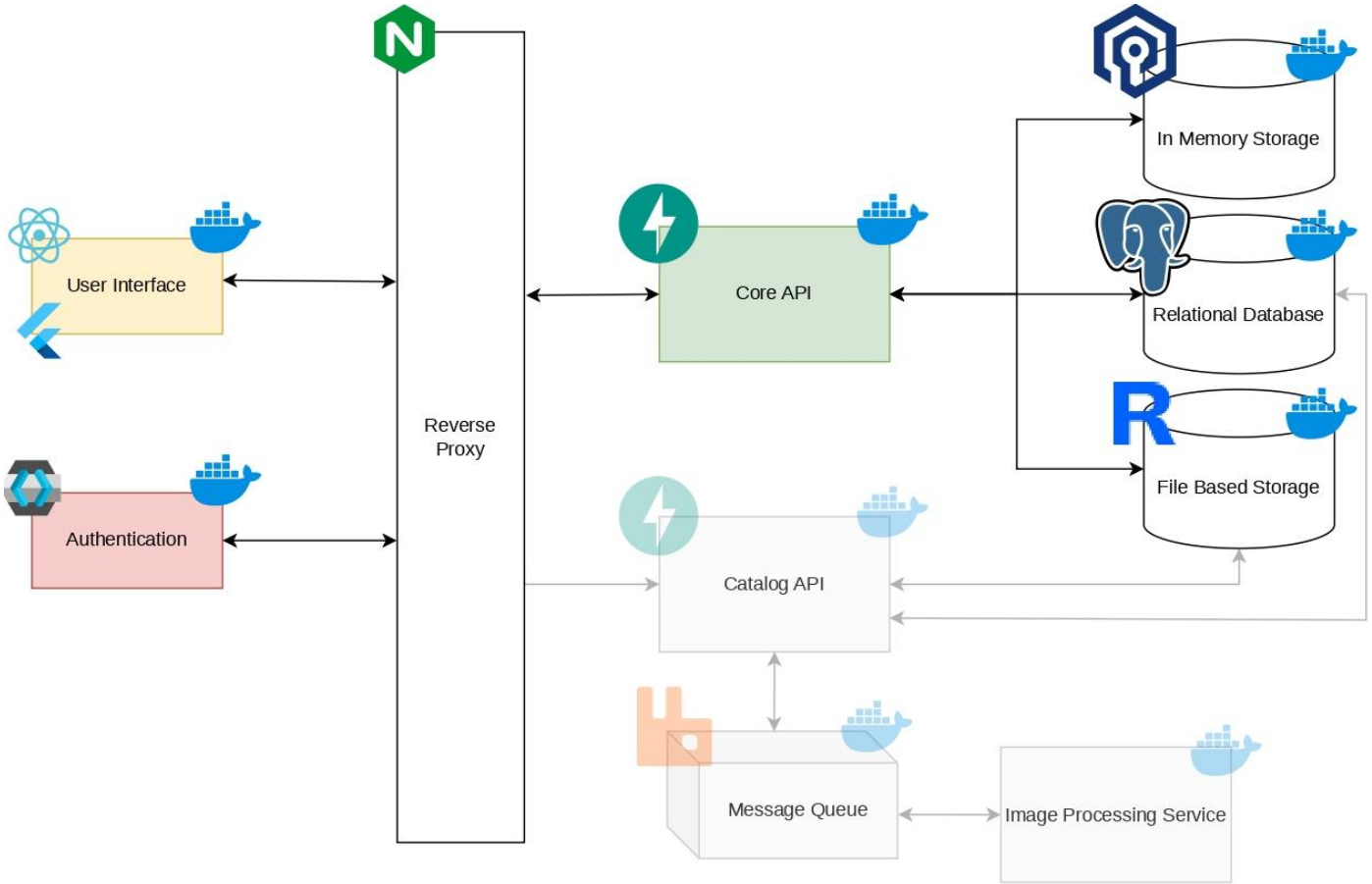




Updated Components Level Architecture


Core API
User Service
CRUD operations for user accounts
Task Service
CRUD operations for tasks
Submissions Service
CRUD operations for tasks
Catalogable Item Service
CRUD operations for Catalogable Items
Inventory View
Retrieves read only historical data from the database
Game Service
Operations for all game related content
Notification Service
CRUD operations for notifications
Social Service
Operations for managing friends
In Memory Storage
(valkey)
Stores high access frequency data
Relational DB
(PostGIS)
Stores content to be displayed in the frontend
File Based Storage
(rustfs)
Stores cataloged images

Updated Container-Level Architecture



Licensing

TreeLens is proprietary.
Rule for every dependency
is that it must not force us
to open our code.



Component	Role	Licence	Compatible with closed code?
NGINX	Reverse proxy	BSD-2-Clause	Yes
Keyclock	Authentication	Apache-2.0	Yes
React	Web front end	MIT	Yes
Flutter	Mobile front end	BSD-3-Clause	Yes
FastAPI	Core API	MIT	Yes
Starlette, Uvicorn	Core API	BSD-3-Clause	Yes
psycopg2	PostgreSQL driver	LGPL-3.0	Yes, if unmodified
PostgreSQL	Relational database	PostgreSQL License	Yes
PostGIS	Geospatial extension	GPL-2.0	Yes, runs as a separate service

Component	Role	License	Compatible with closed code?
Valkey	In-memory store	BSD-3-Clause	Yes
RustFS	File storage	Apache-2.0	Yes
RabbitMQ	Message queue	MPL-2.0	Yes, if its files are unmodified
PyTorch, OpenCV	Image processing	BSD-3-Clause, Apache-2.0	Yes
YOLOx	Tree detection	Apache-2.0	Yes, pending formal check of the pre-trained weights
Blurring model	Faces and plates	-	-
Astro, AstroWind, Tailwind	Documentation site	MIT	Yes

Sector-Specific Regulation



Regulations



AI Act

Minimal risk

- Detects to blur
- Never identifies
- Human validation



DSA

Applies

- Hosting service
- Report button
- Extra duties exempt



INSPIRE

Export ready

- GeoJSON, GeoPackage
- EPSG:3763 coords
- No personal data

AI Act (Reg. 2024/1689) Art. 3(35) · DSA (Reg. 2022/2065) Arts. 16, 19 · INSPIRE (Dir. 2007/2/EC, DL 180/2009)

RGPD

Data protection for the user and for the photo



Personal data

Data	Purpose	Legal basis
Email, password	Account, login	Contract
Display name	Leaderboard, friends	Contract
Age 13+	Minimum age check	Contract
Photos, videos	Trees, Model training	Contract, legitimate interest
GPS or map pin	Tree location	Contract
Gamification data	Points, leagues	Contract
Push token	Notifications	Consent
Access logs	Security, 90 days	Legitimate interest



Data controller

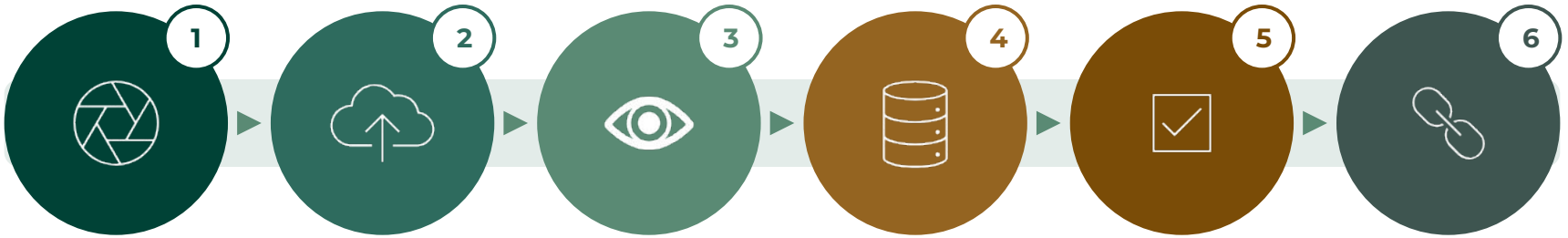
Only the TreeLens team as access do the personal data and no other partner

Right to be forgotten

1. User taps "Delete account"
2. Personal data erased within 30 days
3. Images unlinked, kept anonymous

Legal bases under GDPR Art. 6(1): contract (b), consent (a), legitimate interest (f). Unlinked images are no longer personal data (Recital 26).

Image flow on our system



Capture **Upload** **Blur** **Store** **Review** **Unlink**

If the user deletes account or data

Terms and Conditions



Who is responsible?

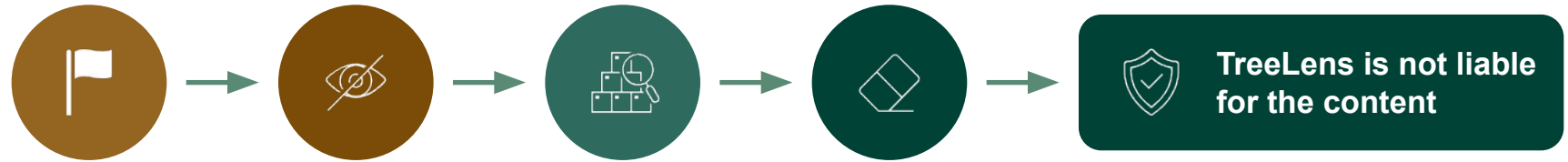
User

Responsible for every submission they make

TreeLens team

Service provider that protects the data and removes illegal images

If an image is illegal, it should be removed from our system:



Using TreeLens



Account

- 13+ years old
- One account per email
- Public or Private



Prohibited use

- Fake location
- Copied photos
- Privacy violation



Image licence

- TreeLens property
- Model training
- Irrevocable



Term changes

- 15 days notice
- Shown in the app
- Can leave first

Breaking the rules results in content removed, points cancelled or account suspended. Terms change only for listed reasons (DL 446/85, Art. 22).

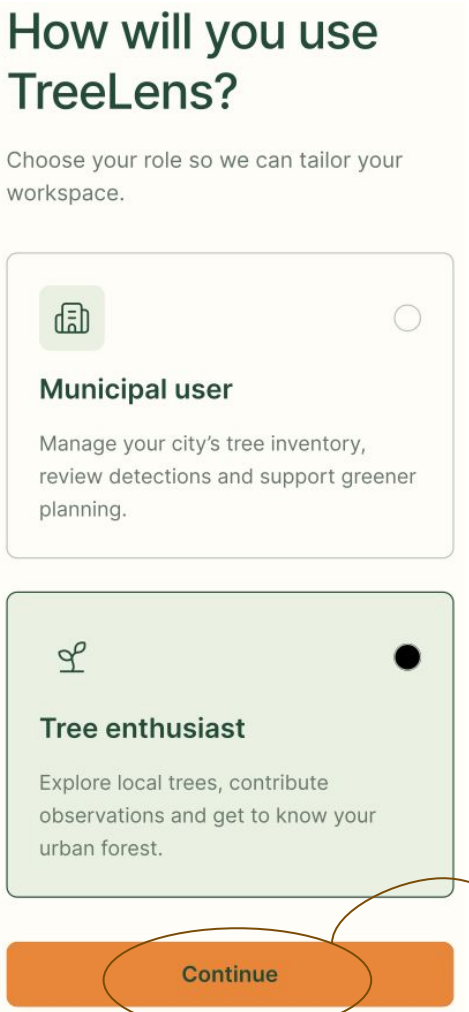
Usability and Digital Accessibility

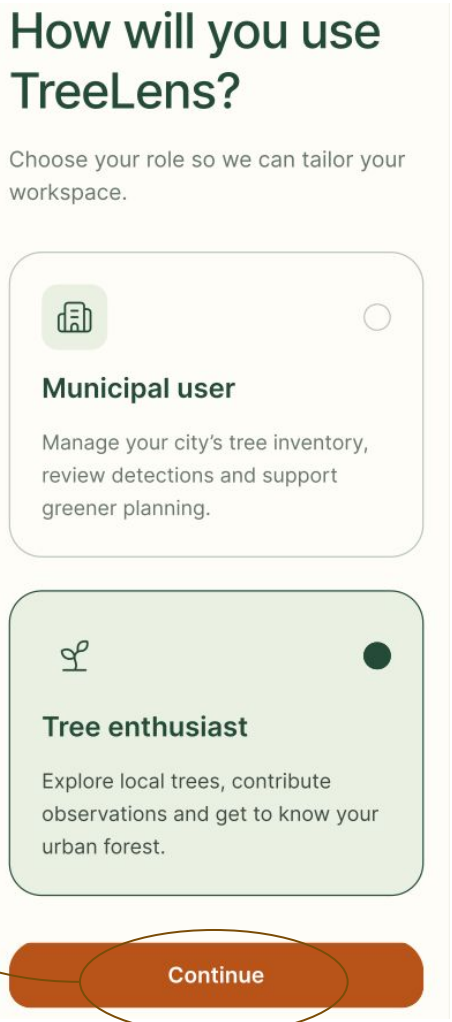
The prototypes developed for MS2 were not accessible to the entire population, issues were identified, and changes were made to them.

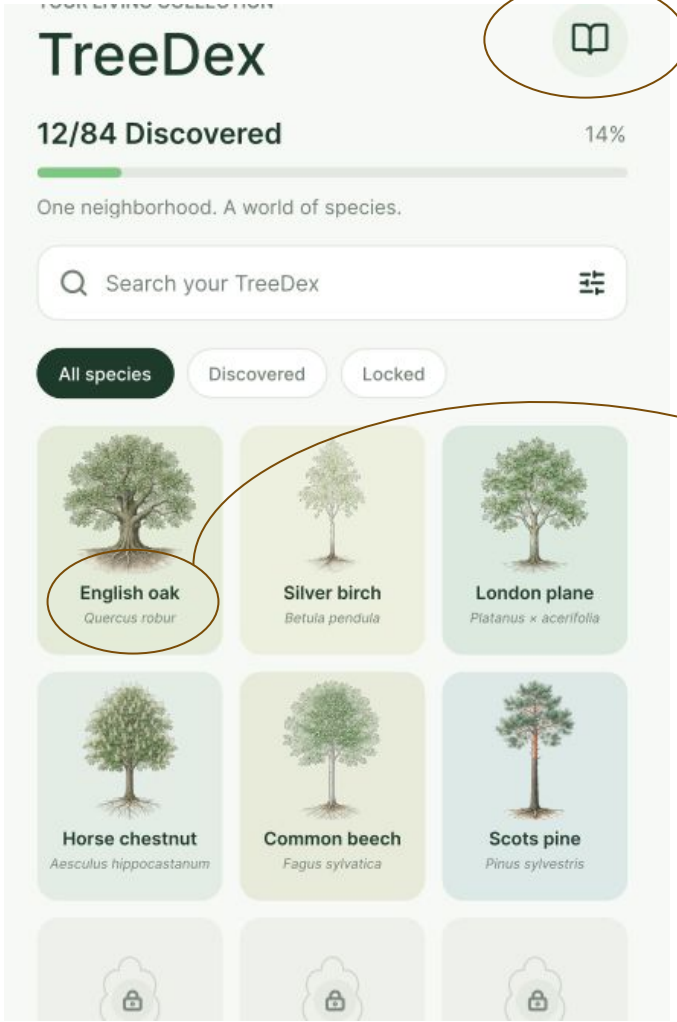


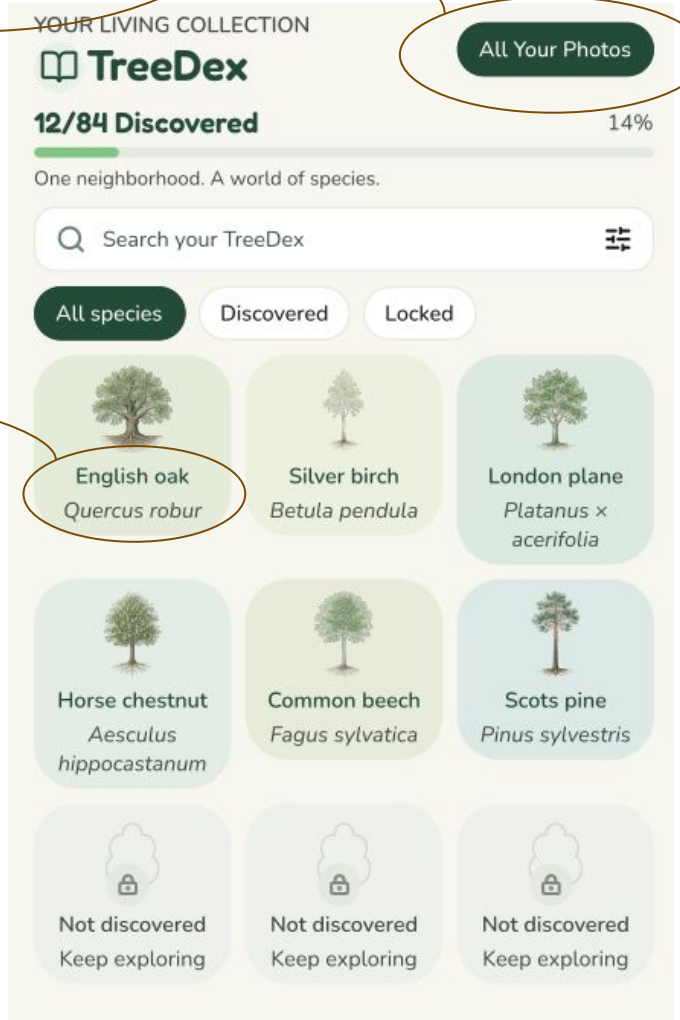
Some Identified Problems

Element	Affected profile	Problem	WCAG	Priority	Correction
Orange buttons (CTA)	Low vision	Dark green text on light orange: insufficient contrast	1.4.3	**High**	Orange #B8541A with white text (contrast >=4.5:1) on all buttons
Secondary text and labels	Low vision	Small, low-contrast text that is hard to read and zoom	1.4.3	**Medium**	Larger text (>12 px), higher contrast, clearer labels
Icon-only buttons	Blind (screen reader users)	Buttons have no text, so a screen reader cannot tell what they do	4.1.2 / 1.1.1	**High**	Accessible name on every icon-only button, plus a visible label or tooltip


How will you use TreeLens?
Choose your role so we can tailor your workspace.
Municipal user
Manage your city's tree inventory, review detections and support greener planning.
Tree enthusiast
Explore local trees, contribute observations and get to know your urban forest.
Continue


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TreeDex
12/84 Discovered
14%
One neighborhood. A world of species.
Search your TreeDex
All species
Discovered
Locked
English oak
Quercus robur
Silver birch
Betula pendula
London plane
Platanus × acerifolia
Horse chestnut
Aesculus hippocastanum
Common beech
Fagus sylvatica
Scots pine
Pinus sylvestris


YOUR LIVING COLLECTION
TreeDex
All Your Photos
12/84 Discovered
14%
One neighborhood. A world of species.
Search your TreeDex
All species
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English oak
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Betula pendula
London plane
Platanus × acerifolia
Horse chestnut
Aesculus hippocastanum
Common beech
Fagus sylvatica
Scots pine
Pinus sylvestris
Not discovered
Keep exploring
Not discovered
Keep exploring
Not discovered
Keep exploring



Keep up with our work and check our calendar for the next milestone!